

Aashish Sinha

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EDUCATION

Indian Institute of Technology (ISM) Dhanbad

Bachelor of Technology in Mechanical Engineering

Dhanbad, India

Nov. 2022 – May 2026

The Senior Study School

High School, Science

Amritsar, India

March 2022

EXPERIENCE

RRC, III-T Hyderabad

Research Intern: Under Prof. Spandan Roy

Feb 2026 – Present

Hyderabad, India

- **Project: Graph-Informed MPC for Drone Swarms**
- Designed a Model Predictive Control (MPC) framework for collaborative multi-drone payload transport.
- Trained a GNN to predict nonlinear MPC residuals and cable tension forces.
- Generalized the GNN across variable swarm sizes and payload weights to mitigate swing and ensure stability.
- **Project: Vision-Guided Aerial Manipulator - Grabbing and Placing**
- Implemented minimum-snap trajectory optimization for smooth quadrotor motion profiles.
- Integrated motion controllers with VLM outputs to execute LiDAR-free autonomous manipulation.
- Stabilized grabbing and placing tasks using the manipulator.

Robotics Lab, IIT Dhanbad

Guided by Prof. Arun Dayal Udai

Nov 2025

Dhanbad, India

- **Project: Humanoid - Grasping with RGB-D Camera**
- Processed Intel RealSense data for 3D pose estimation of an object.
- Solved IK utilizing the robot's URDF and executed bottle-grasping task.
- **Project: Cart-Pole Stabilization System**
- Designed an energy-based control algorithm to dynamically swing a resting pendulum up into a near-upright position.
- Implemented an LQR balancing controller to catch and stabilize the pendulum at the top.

Automation Lab, IIT Dhanbad

Research Assistant: Under Prof. Vivek Bajpai

Aug 2025 – Apr 2026

Dhanbad, India

- **Project: Vision-Guided Mechatronic Gantry System**
- Designed a 3-DOF mechatronic crane and parallelogram gripper, applying spatial kinematics for industrial payload manipulation.
- Fine-tuned a real-time YOLO vision pipeline, executing perspective transforms to map 2D camera feeds into 3D world coordinates.
- Deployed a REST API on an ESP32 edge node to wirelessly synchronize AI perception with physical hardware actuation.

DG Takano

Mechanical Research Intern

May 2025 – Jul 2025

Osaka, Japan

- **Project: Precision Mechanical Actuation & Structural Design**
- Engineered a 3-position mechanical actuation system for variable fluid control using advanced CAD and rapid prototyping.
- Optimized the magnetic coupling interface, increasing retention force by 300% to ensure secure modular docking.
- Applied Design for Manufacturing (DFM) to integrate a sustainability feature, cutting unit production costs by ¥12,000 JPY.

NVCTI (Institute Innovation Hub)

Mentored by Dr. Ashish Siddharth

Jan 2025 – Apr 2025

Dhanbad, India

- **Project: LLM-Integrated Autonomous Mobile Robot (AMR)**
- Designed an AMR chassis in CAD and deployed ROS 2 Nav2 with SLAM with LiDAR for real-time indoor mapping and navigation.
- Fused wheel odometry, IMU, and magnetometer data via Kalman Filter for robust state estimation on a differential-drive prototype.
- Integrated the Gemini LLM API with voice recognition and TTS for natural language-commanded robot control.

PROJECTS

GAN [link] | *Python, PyTorch, Deep Learning*

- Trained a GAN on 60K anime faces.
- Fine-tuned the model for realism in facial features.

ChatPDF [link] | *LangChain, Gemini API, RAG*

- Built a RAG pipeline using LangChain and Gemini API to query online PDFs.
- Bypassed LLM context limits via vector retrieval for massive document Q&A.

TECHNICAL SKILLS

Programming: Python, C++, Linux, Git, REST APIs (FastAPI/Flask)

Robotics & Simulation: ROS/ROS2, Gazebo, PX4

AI & Machine Learning: VLM, PyTorch, GNN, GAN, CNN, Deep RL

Control & Perception: OpenCV, YOLO, SLAM (Nav2), MPC, Kinematics, Trajectory Optimization, Kalman Filters

Hardware & Prototyping: Raspberry Pi, ESP32/Arduino, IMU, 3D Printing, CAD, DFM

RELEVANT COURSEWORK

Robotics & Control: Kinematics & Dynamics of Machinery, PID/MPC, Mechatronics

AI & Computing: Computer Vision, Reinforcement Learning, Data Structures & Algorithms, Embedded Systems

Mathematics: Linear Algebra, Optimization Techniques, Probability & Statistics